

DiaTool-DPO

Enhancing Tool-Augmented LLM Dialogue
via Direct Preference Optimization

Automatic preference pair construction via MDP-based
dialogue state modeling for slot-filling & tool call rejection

arxiv.org/abs/2504.02882

SFT Alone Fails at Slot-Filling & Tool Rejection

Tool-Augmented LLMs must decide: ask follow-up questions, make tool calls, or reject inappropriate requests. SFT with expert trajectories alone cannot reliably learn these dialogue control skills.

Missing Follow-ups

Model skips slot-filling questions when user query lacks required arguments

Hallucinated Tool Calls

Model invents tool parameters instead of asking the user for missing information

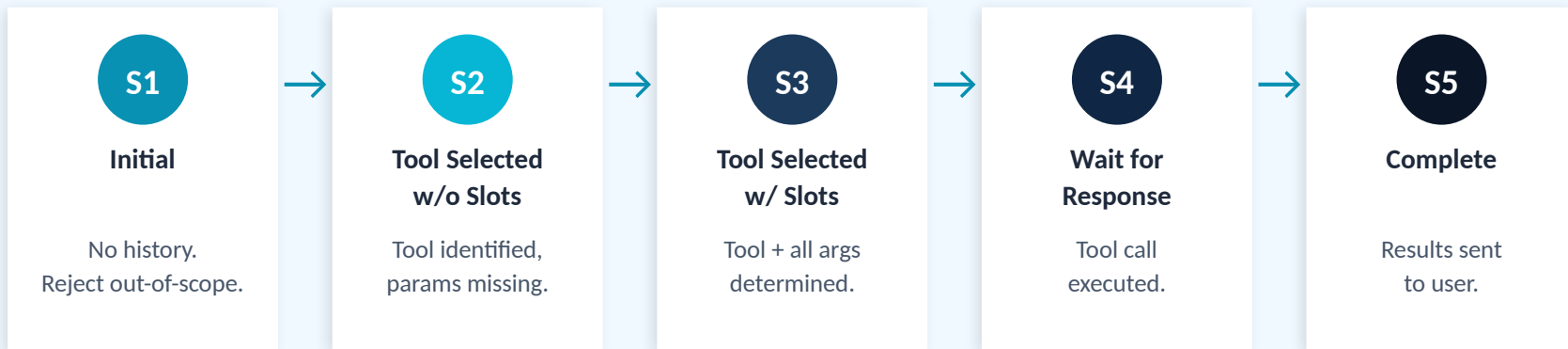
Failed Rejections

Model attempts tool calls for out-of-scope requests instead of declining gracefully



DiaTool-DPO proposes Direct Preference Optimization with automatically constructed paired trajectories — teaching the model which dialogue flows are correct vs. incorrect without human labeling.

Modeling TA-LLM Dialogue as a 5-State MDP



↪ Reject (Type 3)

↪ Slot-filling loop (N times)

State Trajectories by Query Type

- Type 1: $S1 \rightarrow S3 \rightarrow S4 \rightarrow S5$
- Type 2: $S1 \rightarrow (S2 \times N) \rightarrow S3 \rightarrow S4 \rightarrow S5$
- Type 3: $S1 \rightarrow S1$

All info provided

Slot-filling needed

Out-of-scope rejection

Three Query Types Define State Trajectories

Query Type	State Trajectory	Description	Example
Type 1 Complete Query	1 → 3 → 4 → 5	All required arguments present. No slot-filling needed.	"Book a flight from NYC to LA on March 15 for 2 passengers"
Type 2 Incomplete Query	1 → (2×N) → 3 → 4 → 5	Missing arguments require N rounds of slot-filling QA.	"Book a flight" → model asks: origin? destination? date?
Type 3 Out-of-Scope	1 → 1	No suitable tool available. Must reject the request.	"What's the meaning of life?" → model: not a tool-supported query



These three trajectory patterns form the basis for automatic preference pair construction — each type generates its own set of chosen (correct) and rejected (incorrect) dialogue flows for DPO training.

Automatic Paired Trajectory Construction

No human labeling required — preference pairs generated from MDP state transitions

Query	Chosen Trajectory	Rejected Trajectory	Learning Lesson	Count	Difficulty
Type 1	1→3→4→5	1→2→3→4→5	Prevent redundant slot-filling	2,089	Easy
Type 1	1→3→4→5	1→(2×N)→3→4→5	Prevent redundant slot-filling	3,092	Hard
Type 1	1→3→4→5	1→1	Tool call accept	2,652	Easy/Hard
Type 2	1→(2×N)→3→4→5	1→3→4→5	Prevent slot hallucination	2,651	Easy/Hard
Type 2	1→(2×N)→3→4→5	1→(2×M)→3→4→5	Prevent slot hallucination	2,530	Hard
Type 3	1→1	1→3→4 / 1→(2×N)→3→4	Tool call reject	1,129	Hard



Chosen trajectories follow correct MDP transitions | Rejected trajectories introduce failure patterns | Zero human annotation

SFT + DiaTool-DPO Achieves Best Macro Average

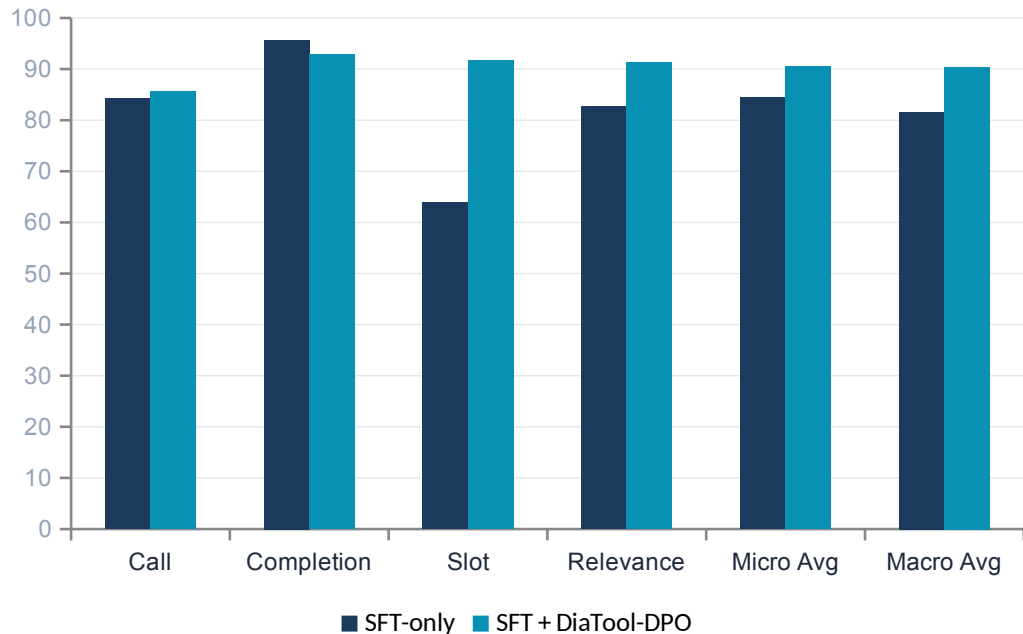
FunctionChat-Bench evaluation across three model architectures

Model	Method	Call	Compl.	Slot	Relev.	Micro	Macro
Prop.-8B	DPO-only	0.314	0.700	0.833	0.609	0.575	0.614
Prop.-8B	SFT-only	0.900	0.916	0.694	0.913	0.870	0.856
Prop.-8B	SFT + DPO	0.886	0.929	0.833	0.826	0.884	0.868
Prop.-3.1B	DPO-only	0.357	0.551	0.528	0.391	0.455	0.457
Prop.-3.1B	SFT-only	0.771	0.817	0.750	0.826	0.790	0.791
Prop.-3.1B	SFT + DPO	0.743	0.871	0.833	0.826	0.765	0.818
LLaMA-3-8B	DPO-only	0.029	0.449	0.056	0.261	0.205	0.199
LLaMA-3-8B	SFT-only	0.843	0.957	0.639	0.826	0.844	0.816
LLaMA-3-8B	SFT + DPO	0.857	0.929	0.917	0.913	0.905	0.904

Green rows = SFT + DiaTool-DPO | Bold teal = best per model | Consistent Macro Avg. improvement across all models

LLaMA-3-8B: +43.5% Slot, +10.5% Relevance

SFT-only vs. SFT + DiaTool-DPO on FunctionChat-Bench



Slot Accuracy

0.639 → **0.917** +43.5%

Relevance

0.826 → **0.913** +10.5%

Macro Average

0.816 → **0.904** +10.8%

Near GPT-4o performance: 94.8% information gathering | 91% tool call rejection

Key Takeaways

RL-Based Dialogue Control for Production TA-LLMs

1

MDP Formulation Enables Systematic Pairing

5 internal states + 3 query types → automatic chosen/rejected trajectory pairs without human annotation.

2

DPO Complements SFT, Not Replaces

DPO-only performs poorly. SFT provides the foundation; DiaTool-DPO adds fine-grained dialogue control.

3

Slot-Filling & Rejection See Largest Gains

LLaMA-3-8B: +43.5% Slot accuracy, +10.5% Relevance — the exact failure modes SFT struggles with.

4

Language-Agnostic Design

Primary experiments in Korean, validated in English. MDP framework generalizes for broad production use.

5

Path to GPT-4o-Level Tool Use on Smaller Models

Open-source 8B models reach near GPT-4o conversational tool-use performance with SFT + DiaTool-DPO.